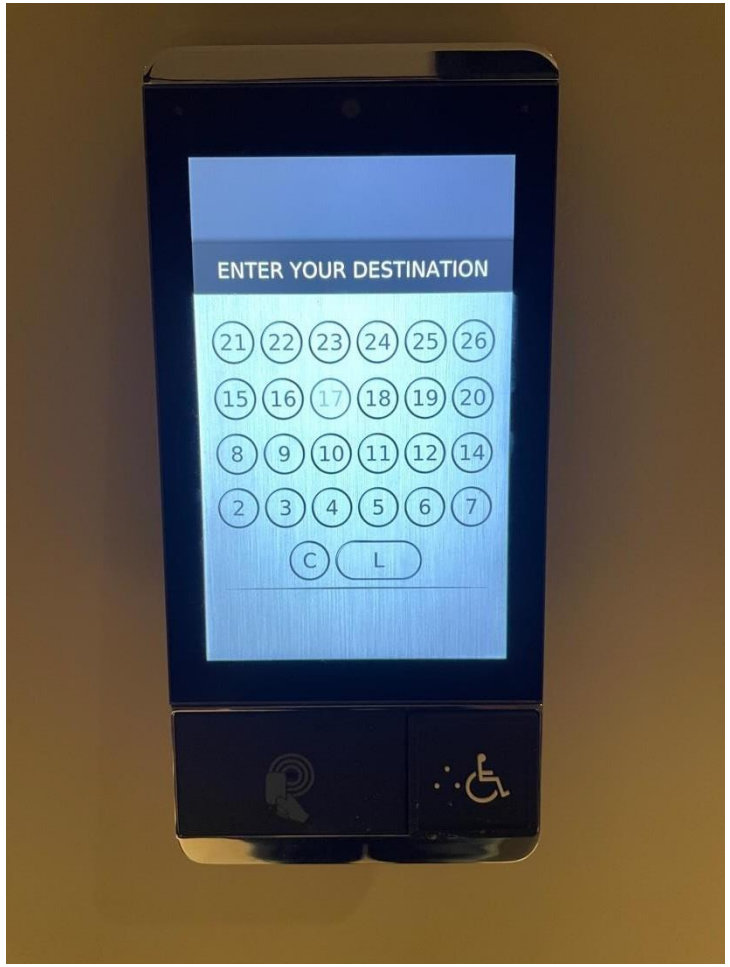


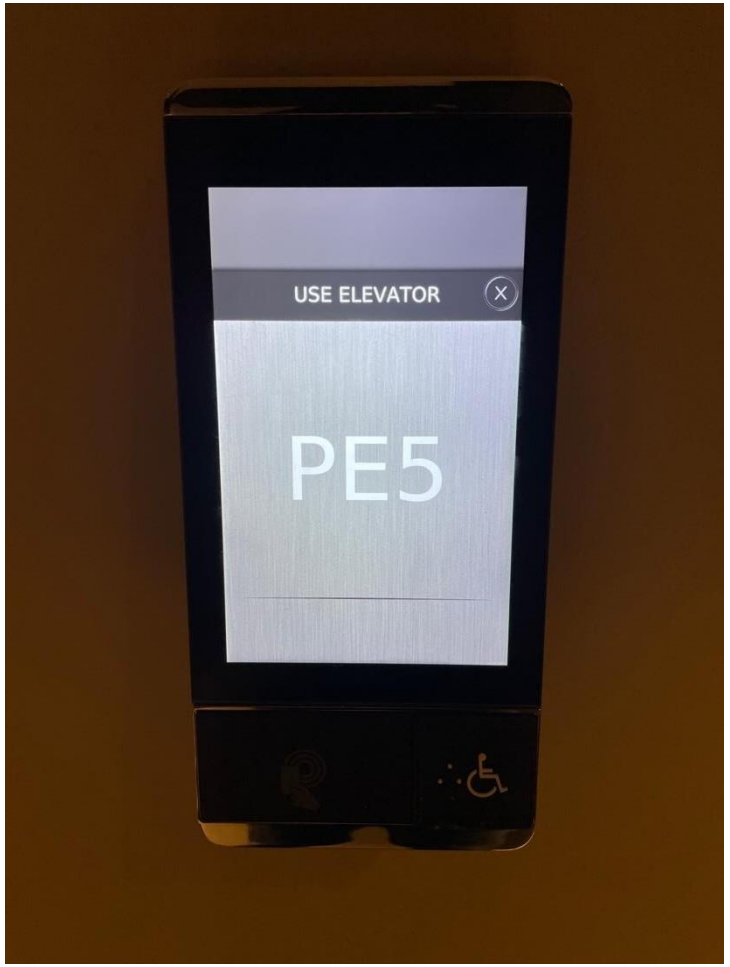
Intro to Prototyping and Low-Fi Prototypes

Human Computer Interaction

Catia Prandi
catia.prandi@unibo.it

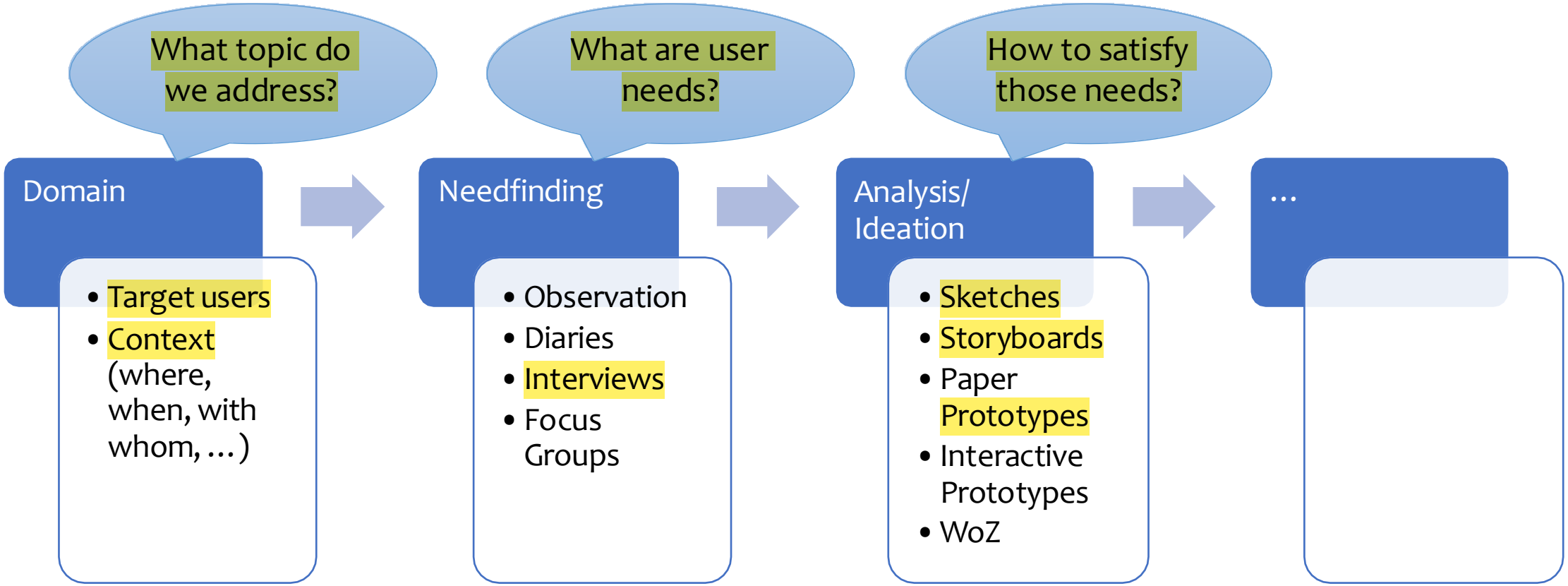
Hall of Fame or Shame?





Process Recap

Result: a solution driven by user needs





Prototype to avoid errors

The Goal

We will see Prototyping

- **Envisionment**: making ideas visible
  - Generating new ideas
  - Evaluating new ideas (within the design group)
  - Testing new ideas (with users) we will not
- Different tools and techniques, according to
  - The stage of design (early, …, advanced, final) 3 types of prototyping: low-level, mid-level and high-level
  - The intended audience (designers, test users, clients, management, …)

according to Design theories and principles and guidelines

- **Error to avoid**: focusing on the user interface before focusing on the task that the user has to accomplish

The Method

- Techniques to **explore** different design **alternatives**
- Explore different solution about different
  - Flows of action
  - Devices and their roles
  - Interfaces
- Alternatives
  - More than one possible design
  - Impossible to get it right the first time
  - Find the best possible solution (not the easier one to implement)

Techniques

- Sketches (see "Storyboards")
- Maps (navigation maps)
- Prototypes:
  - Low Fidelity (paper)
  - Video we will not use them
  - Medium Fidelity (digitally, but not the final one)
  - High Fidelity (using Figma. this is the final interface)

"If a picture is worth a thousand words, a prototype is worth a thousand meetings" — IDEO

Maps

Visual overviews of navigation paths

Navigation Map

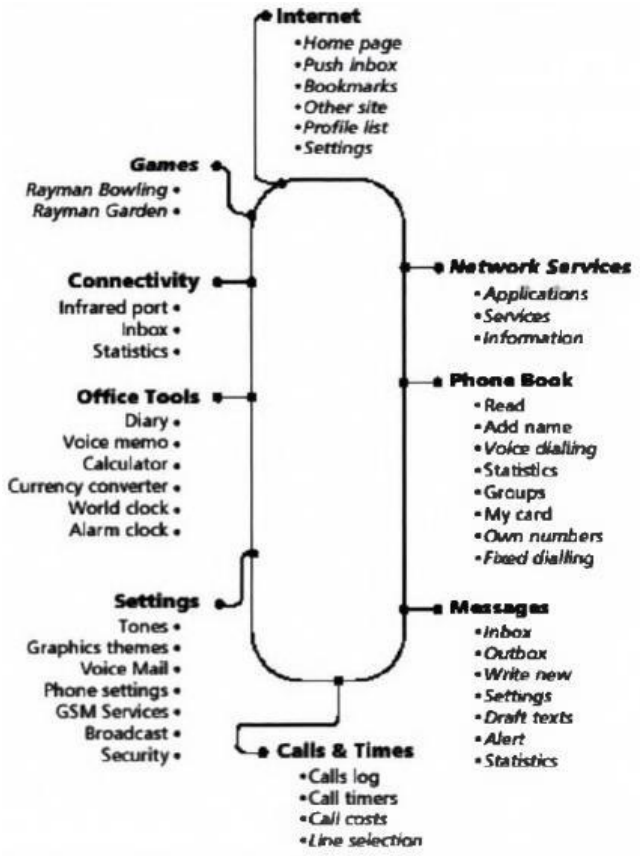
a way to create different connection between different areas/interfaces

- A high-level view for the major structure of the interface
- Focus on how people move throughout the application
- Does not show the pages, only their organization and hierarchical relationship
- Related to the "information architecture" of the application (with map we are structuring information)

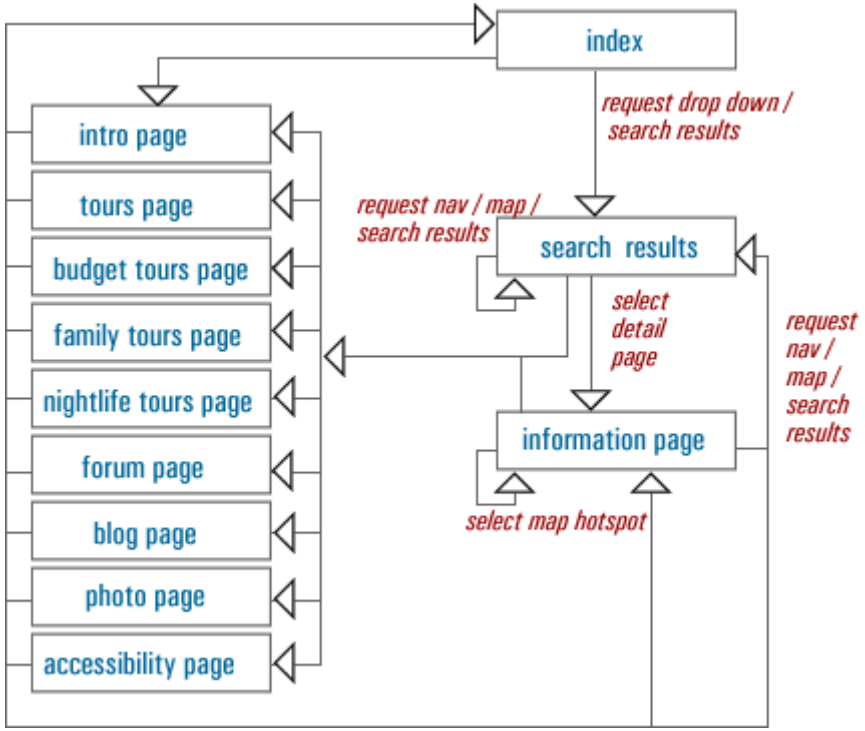
Map Examples

Mental map about the app organisation between interfaces

Old-style mobile phone menus



Website 'sitemap'



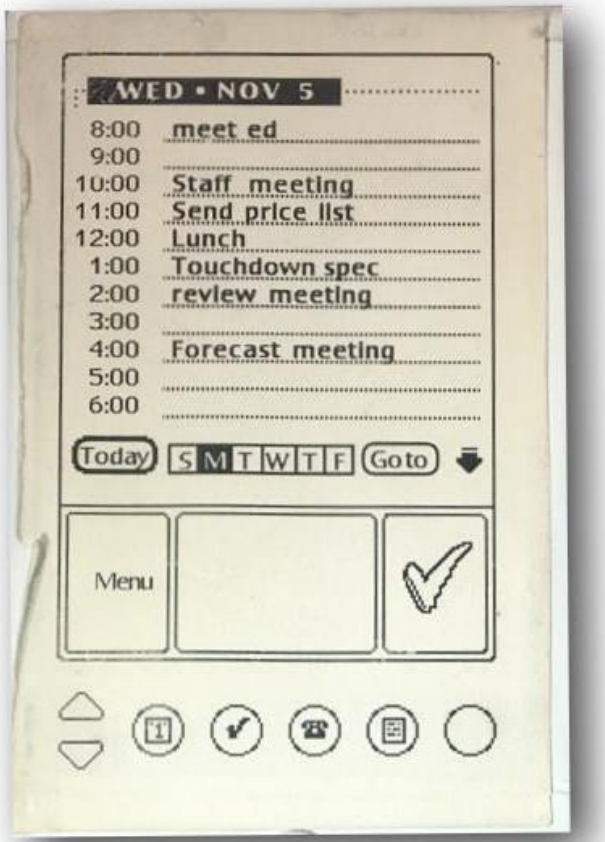
Prototypes

Tangible or digital approximations, at various levels, of system behavior and appearance, to cheaply and quickly evaluate and explore design decisions

Prototypes

- «A prototype is a concrete but partial representation or implementation of a system design»
- «An easily modified and extensible model (representation, simulation or demonstration) of a planned software system, likely including its interface and input/output functionality» easy to add/modify/remove functionalities/aspects of prototype
- One of the most powerful tools for design exploration, visualization, and testing
- They let us 'see' and 'feel' interactivity (simulated or real)

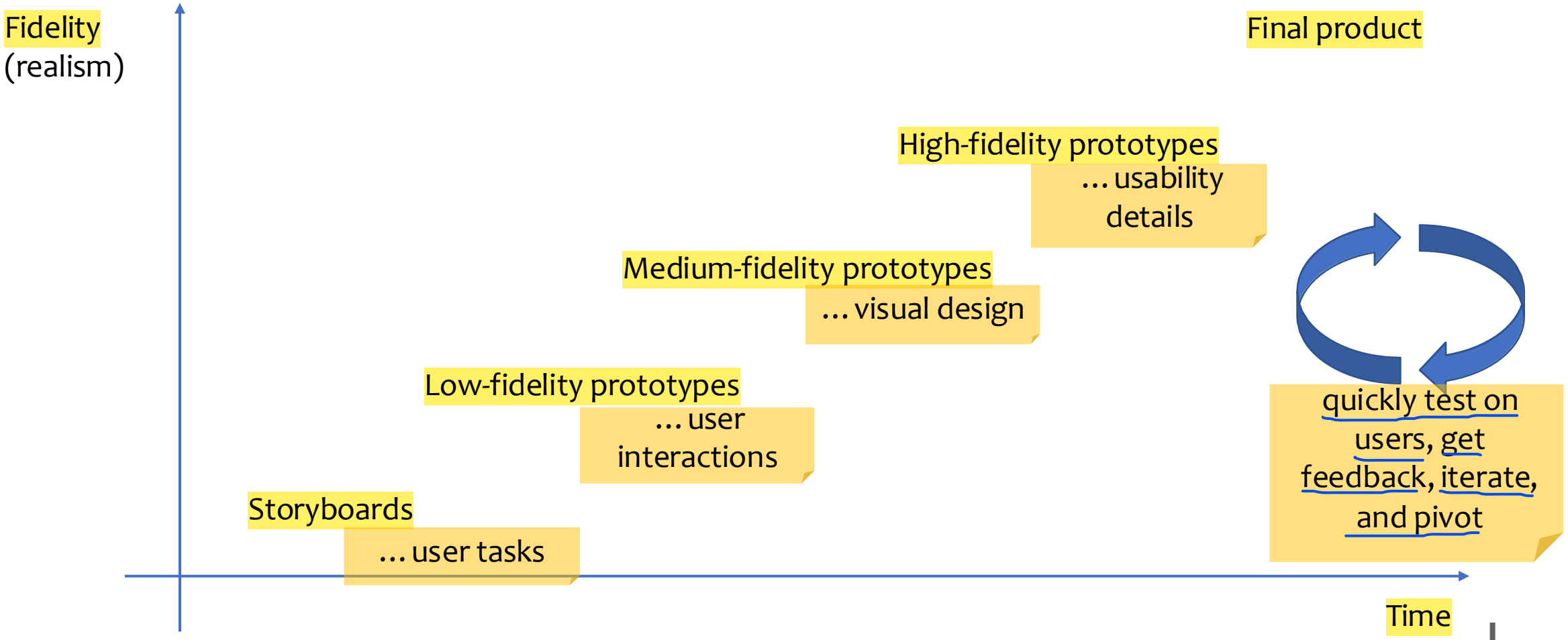






source: https://albertosavoia.medium.com/the-palm-pilot-story-1a3424d2ffe4

Prototypes Facilitate Conversations About…



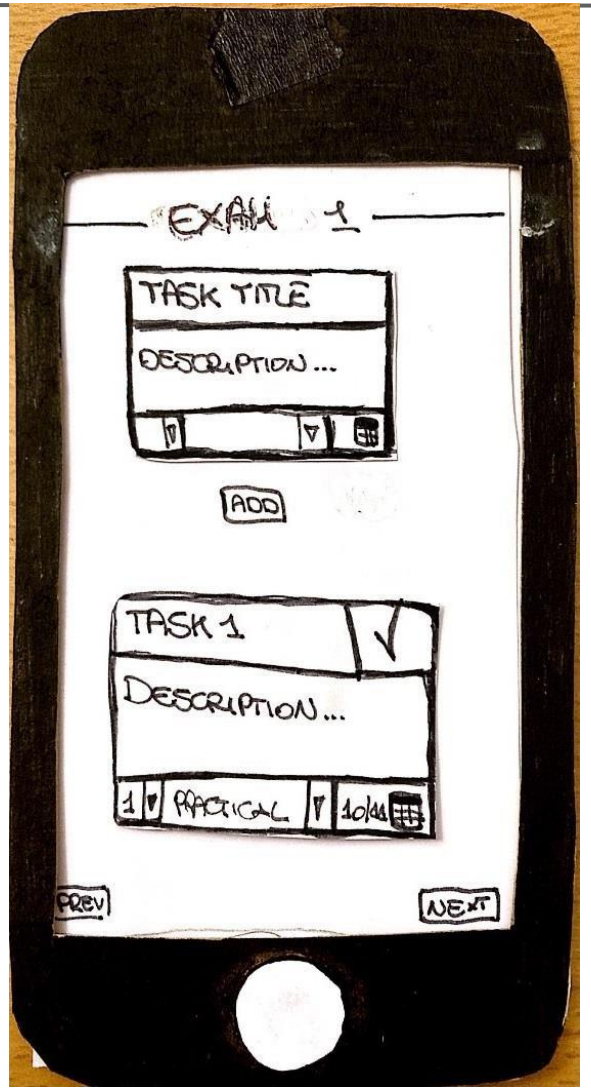
Low to High Fidelity Prototypes

not all of Design principles can be applied on LowFidelity Prototypes (but most of them yes)

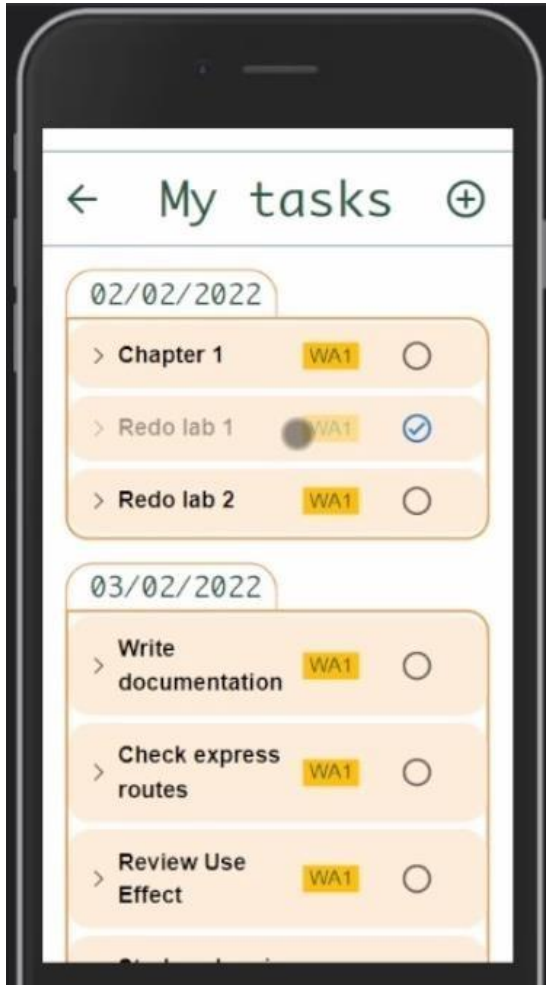
Low-fi

- Lays out the *main* information, interactions, and design choices
- With many missing details

(e.g. Color principles not applied, but Consistency principle yes)



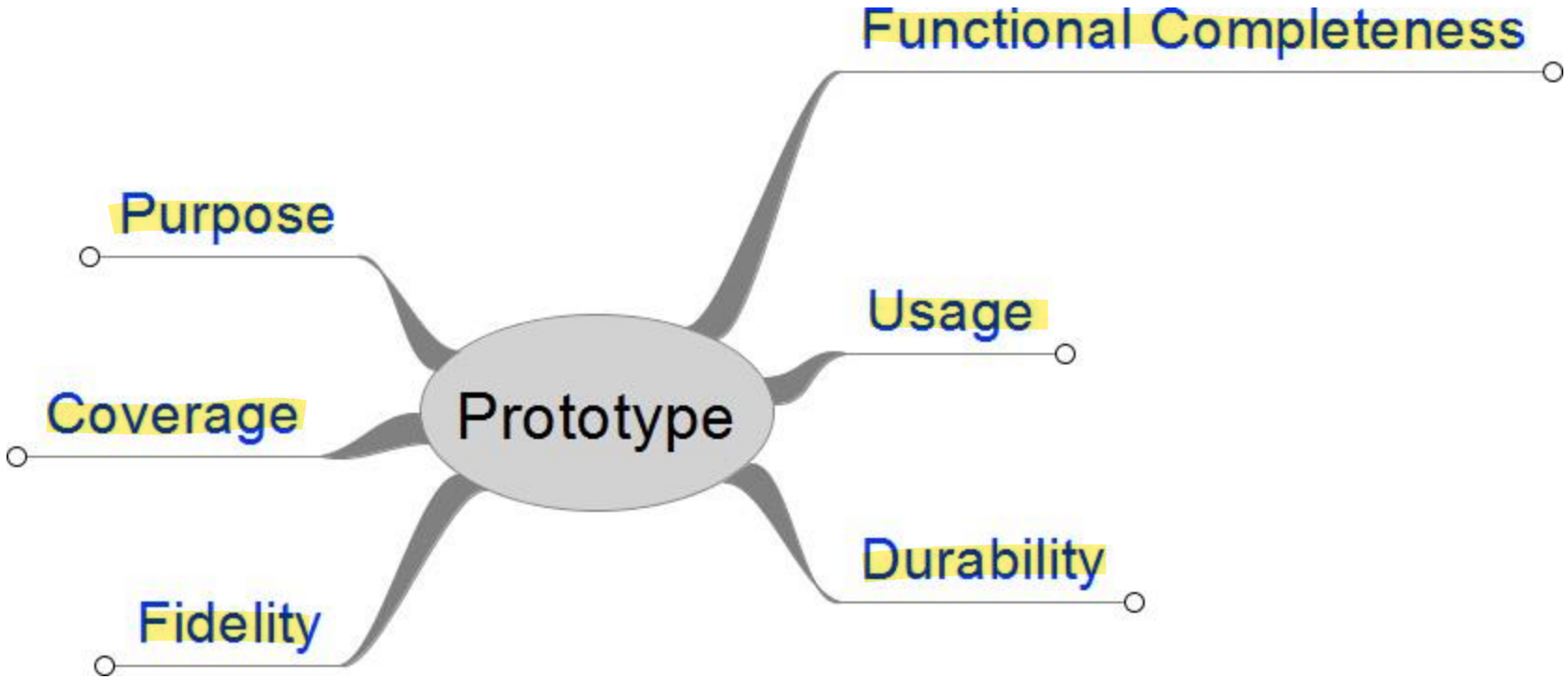
Low to High Fidelity Prototypes



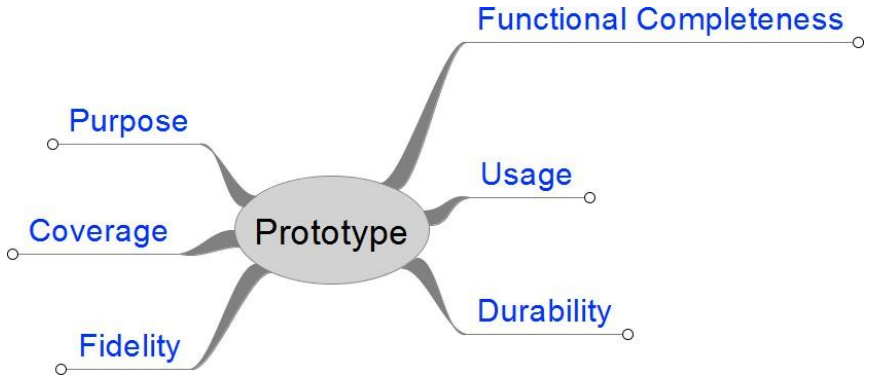
Hi-fi

- It *looks like* the final product

Characteristics of Prototypes



Characteristics of Prototypes



How the system will be used by user?

To evaluate the role of a product in the user's life

Role

How to create and let the user interact with it?

To evaluate interaction modality between user and product

Interface

3 types of Purpose

Purpose

Start think to tech aspects

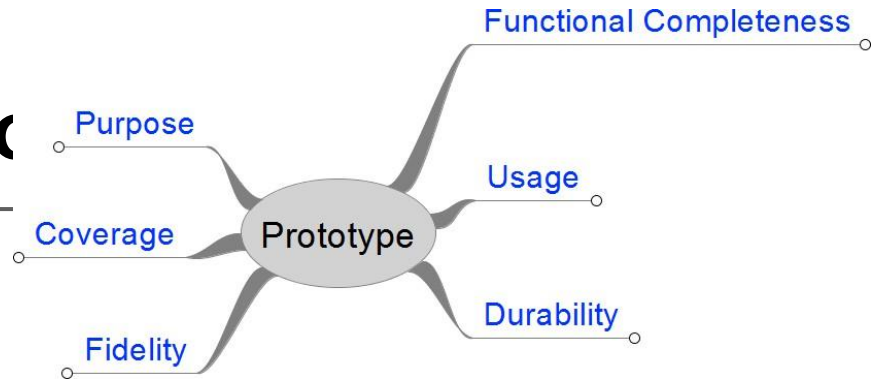
To evaluate technical aspects of product realization

Implementation

Possible Purposes For a Prototype

- Expert analysis
- Check with design rules and guidelines
- Involve users in a controlled experiment
- Involve users “in the wild”
- …

Characteristics of Prot



Not so durable (changes are the normal)

3 types of Durability

Durability

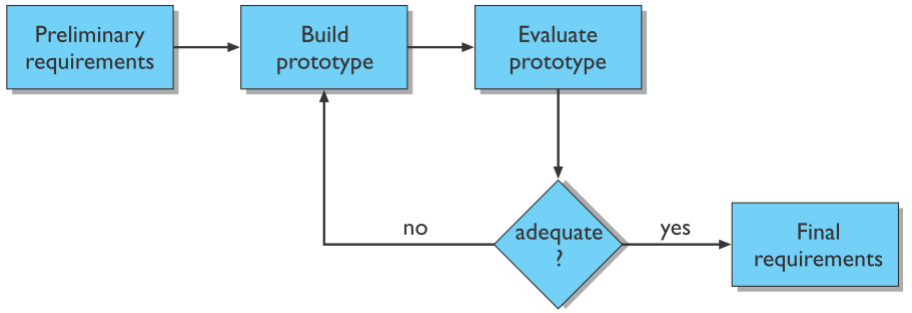
- **Exploratory** — A throw-away prototype used to clarify project goals, to identify requirements, to examine alternative designs, or to investigate a large and complex system
- **Experimental** — A prototype used to validate system specifications (create various type of interface to experiment and evaluate)
- **Operational** — An iterative prototype that is progressively refined until it becomes the final system

Show the interactions in prototype

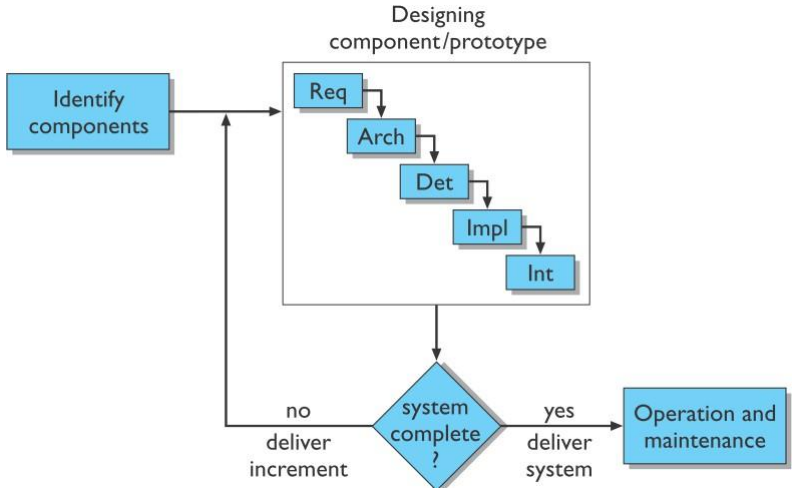
Durability (1)

Three types of prototype

- **Throw-away prototype**: used to assess some qualities of the system (gain knowledge), then discarded

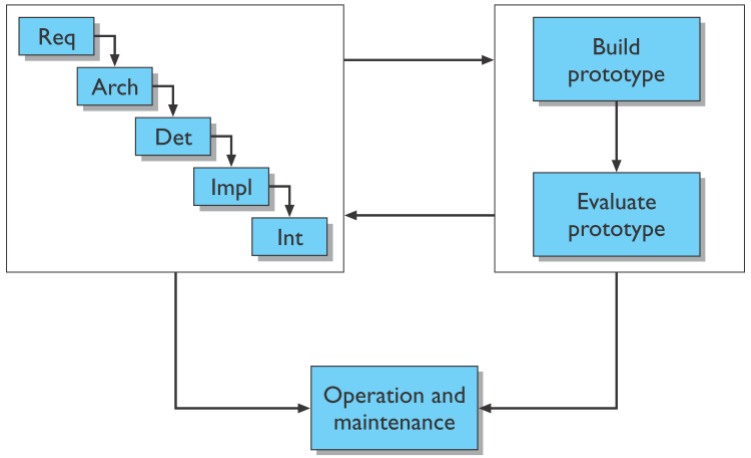


- **Incremental prototype**: the system is developed as incremental modules, each of them released in a separate step

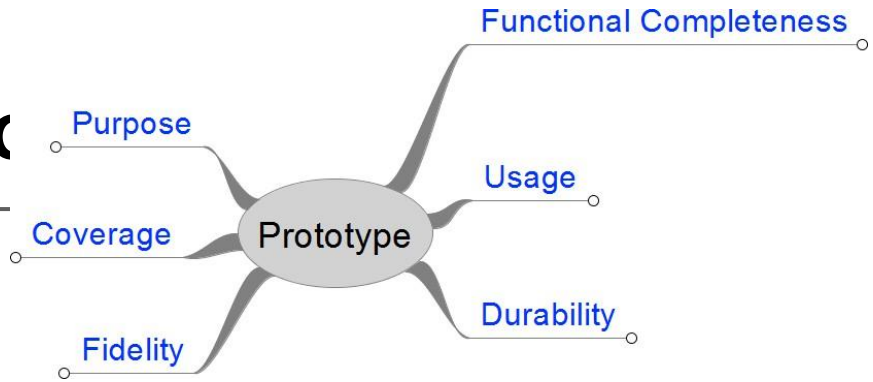


Durability (2)

- **Evolutionary prototype**: the prototype *becomes* the product; each product iteration builds upon the previous one



Characteristics of Prot



A prototype of the entire system

- an expanded horizontal prototype
- models a greater number of features
- covers multiple levels of the system's structure chart
- useful throughout the design process

Global

2 types of Coverage

Coverage

A prototype of a single usability-critical system component

- a vertical prototype that is focused on one feature
- useful at some specific stage of the design process

Local

A prototype that models many features but with little detail

- a horizontal slice of a system's structure chart from thetop down to a specific depth
- most useful in the early stages of design
- purpose is to test the overall interaction metaphor, so includes common functions that the user is expected to perform frequently

Horizontal

3 types of Functional Completeness

Functional Completeness

A prototype that models few features but with much detail

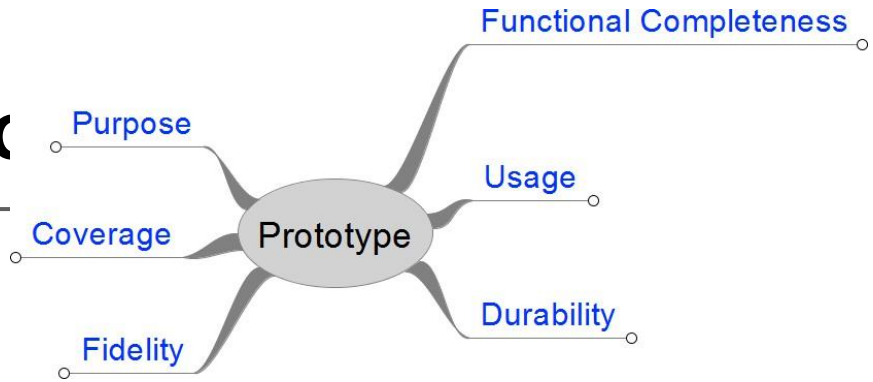
- a vertical slice of a system's structure chart from top to bottom
- most useful in the later stages of design
- purpose is to test details of the design

Vertical

Diagonal

A prototype that is horizontal down to a particular level, then vertical below that point

Characteristics of Proto



A set of drawings (e.g., storyboard) that provide a static, non-computerized, non-working mock-up of user interface for the planned system

A set of screens that provide a dynamic, computerized, working model of the planned system

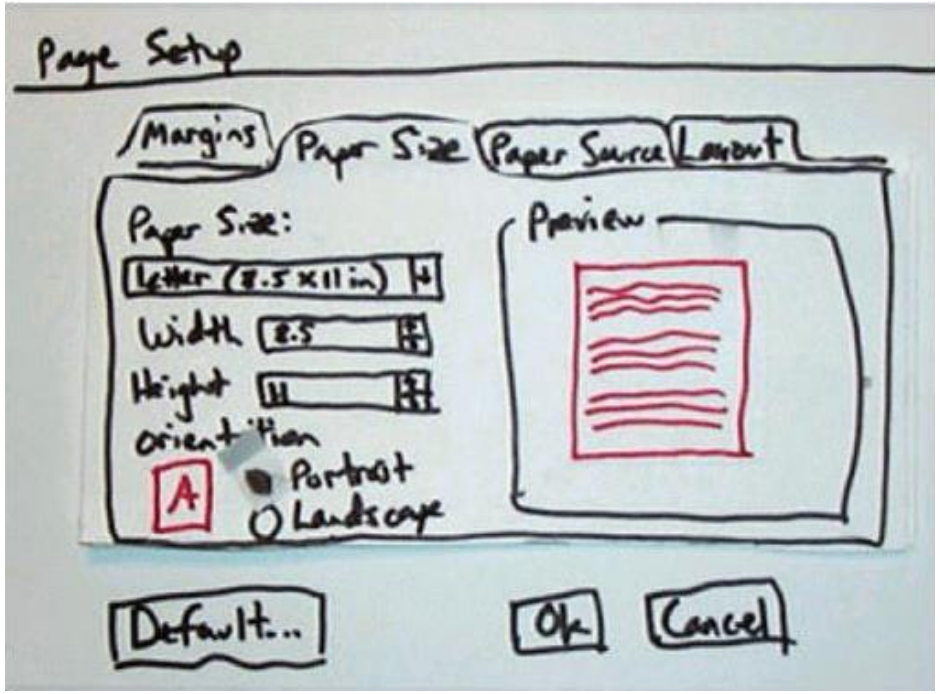
3 types of Fidelity

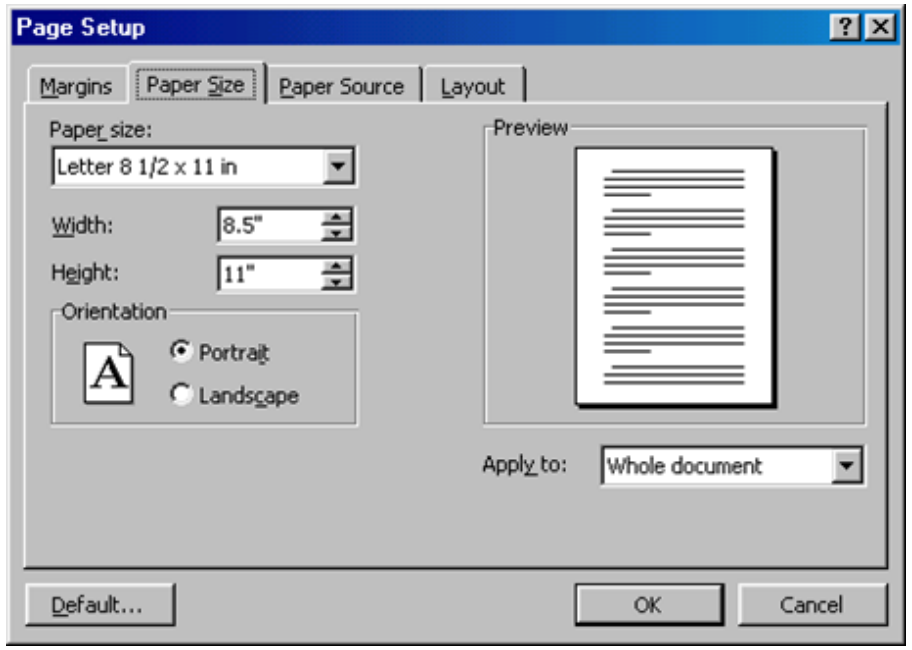
- Fidelity
  - Low
  - Medium
  - High

Our LowFi will be Dynamic

- Usage
  - Static: Static representation of the product (storyboards, diagrams, ...)
  - Dynamic: Dynamic (but not interactive) representation of the product (e.g., video)
  - Interactive: Allows users to test the usage of the system, even if in an approximate and simplified way

Fidelity: Different Information Is Conveyed





Low Fidelity Prototypes

How to start using an application, months before implementing it

Paper Prototypes

- A hand-drawn mock-up of the user interface (usually) on multiple sheets of paper of varying sizes



Key Features for Paper Prototypes

- Interactive paper mockup
  - Sketches of screen appearance
  - Paper pieces show windows, menus, dialog boxes
- Interaction is natural
  - Pointing with a finger = mouse click
  - Writing = typing
- A person simulates the computer's operation (simulate all the interactions)
  - Putting down & picking up pieces
  - Writing responses on the "screen"
  - Describing effects that are hard to show on paper (don't speak during showing prototype)
- Low fidelity in look & feel (simple UI)
- High fidelity in depth (person simulates the backend)

- http://web.mit.edu/6.813/www/sp18/classes/11-prototyping/



Colors only to emphasise (not used normally)

Materials

- Paper, Transparent paper
- Pens, Markers
- Post-It notes
- Glues, scotch tape, scissors
- Photocopies
- UI Stencils
- Reusable UI components
- Printouts of screenshots



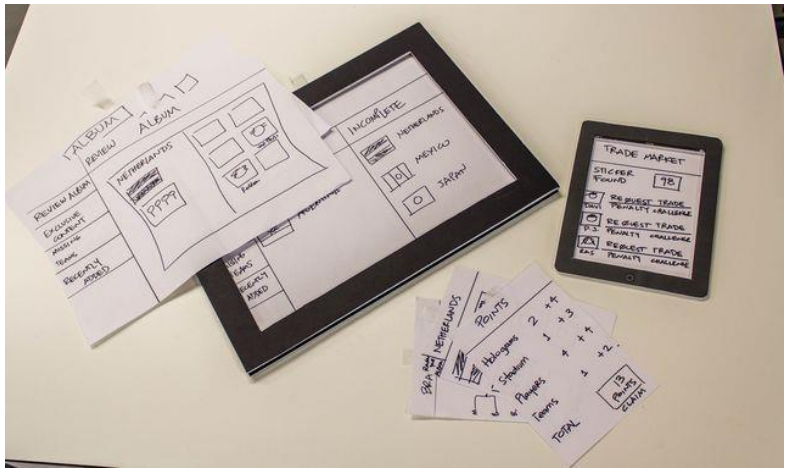
Why Paper Prototyping?

- Faster to build
  - Sketching is faster than programming
- Easier to change
  - Easy to make changes between user tests, or even **during** a user test
  - No code investment - everything will be thrown away (except the design)
- Focuses attention on big picture
  - Designer doesn't waste time on details
  - Customer makes more creative suggestions, not nitpicking
- Non-programmers can help
  - Only kindergarten skills are required

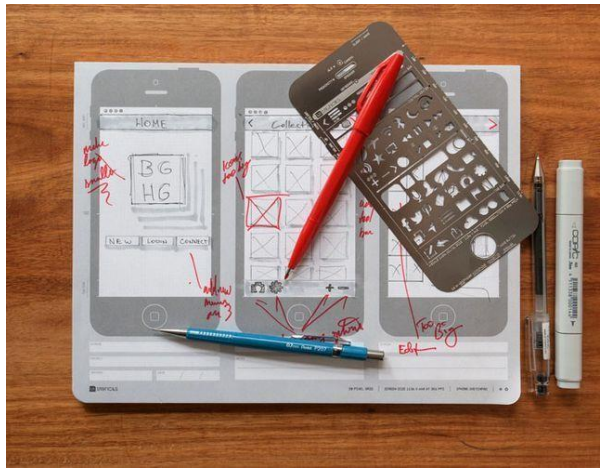
- http://web.mit.edu/6.813/www/sp18/classes/11-prototyping/

Paper Prototypes: Examples

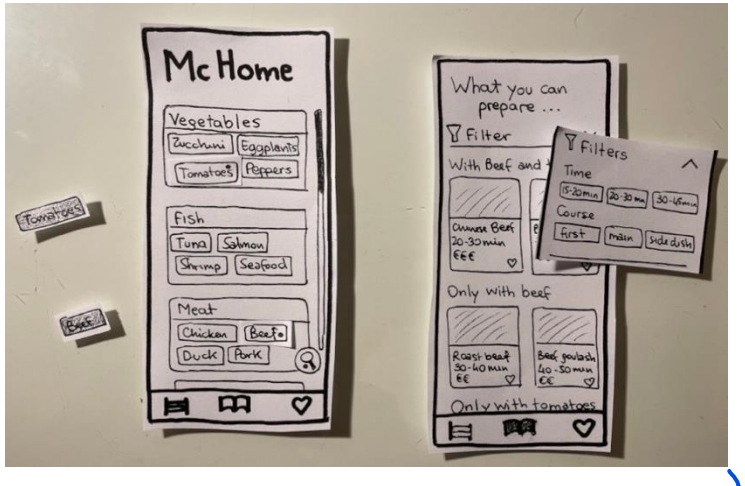
Foldable and Tablet (two type of prototype)



Print the shape of smartphone and then draw



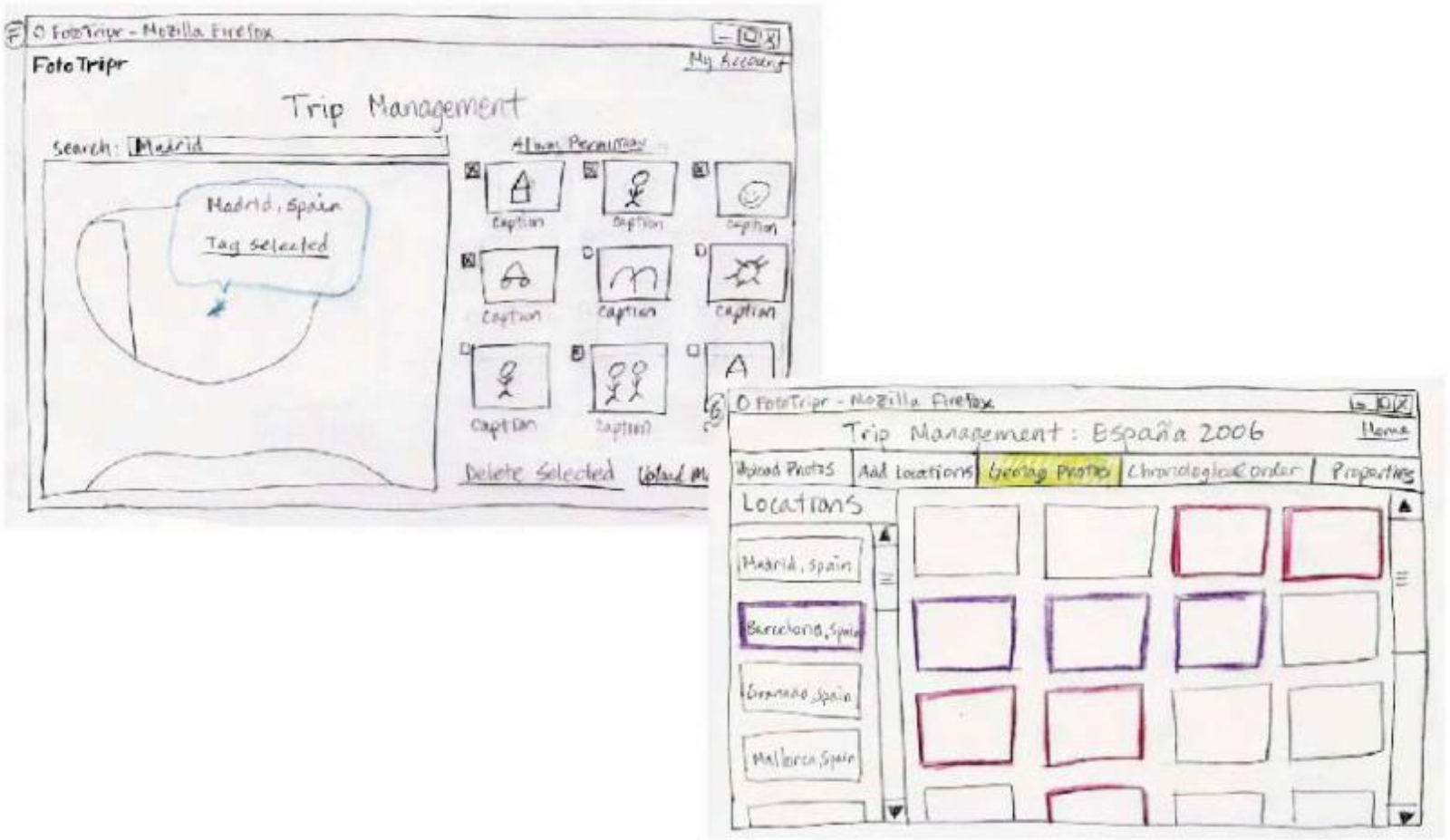
Not so simple to
manage during demo



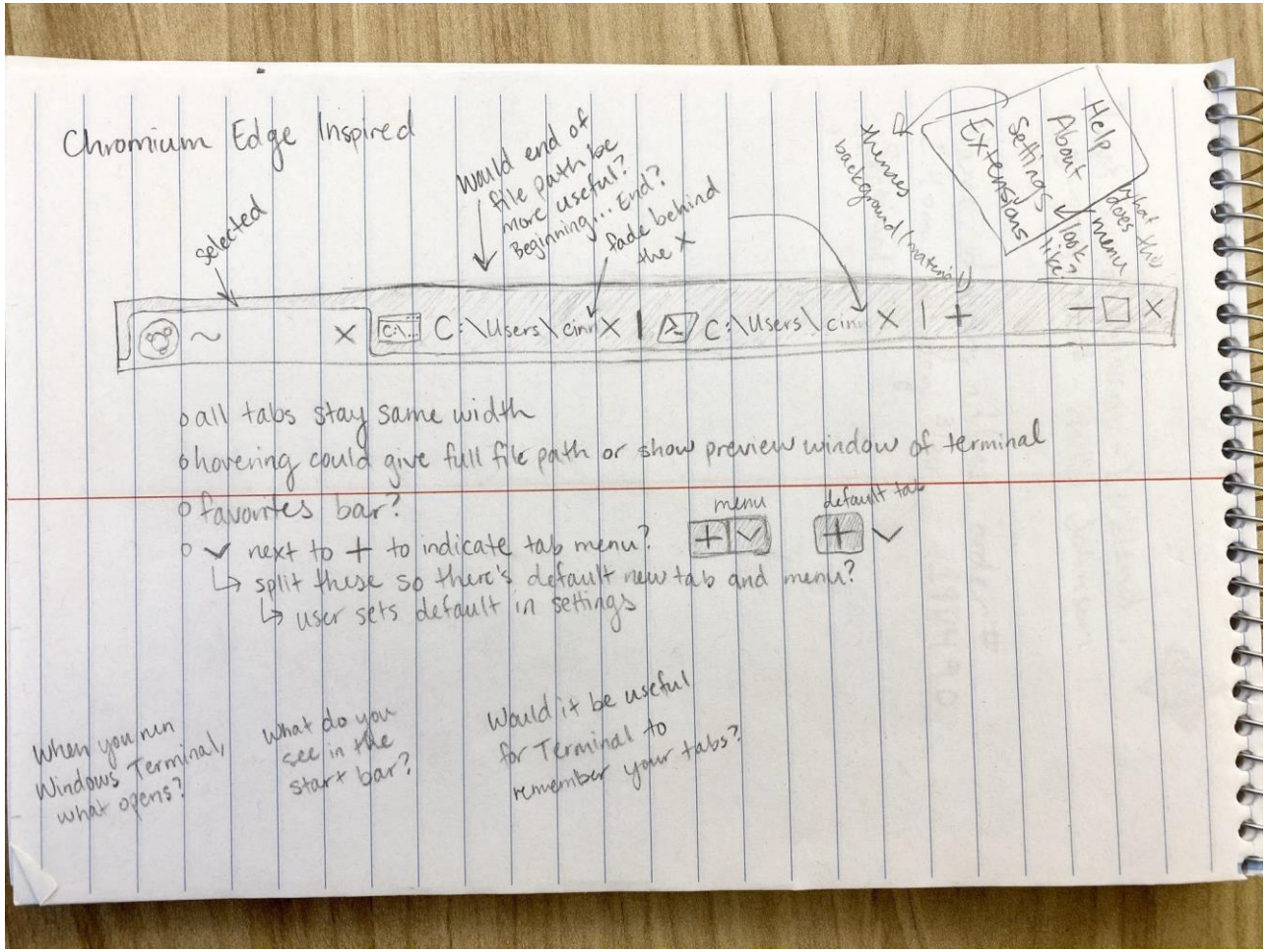
(e.g. open filters and put the
filters subpart on the paper)

Other Examples

Don't use colors too much



First Ever Mockup of the Windows Terminal Tab Bar



https://twitter.com/cinnamon_msft/status/1190015862201176065?s=20

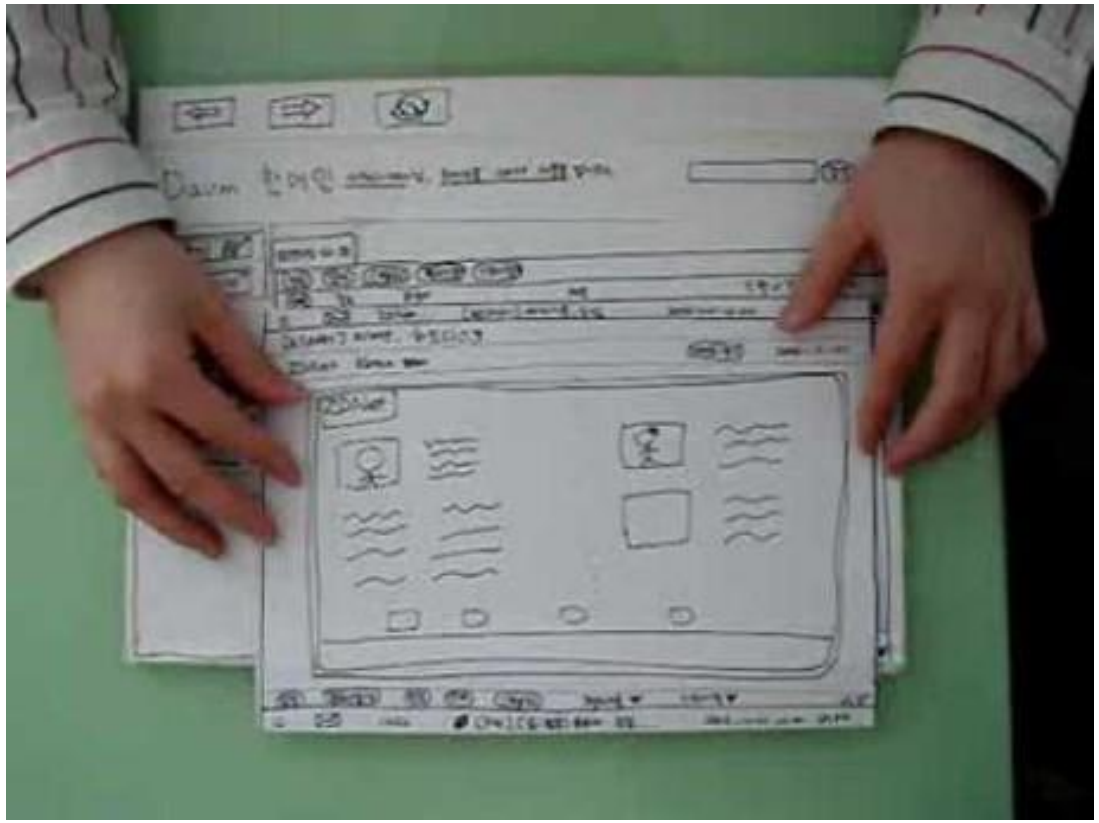
for specific tasks

Creating Flows With Paper Prototypes

A lot of screens



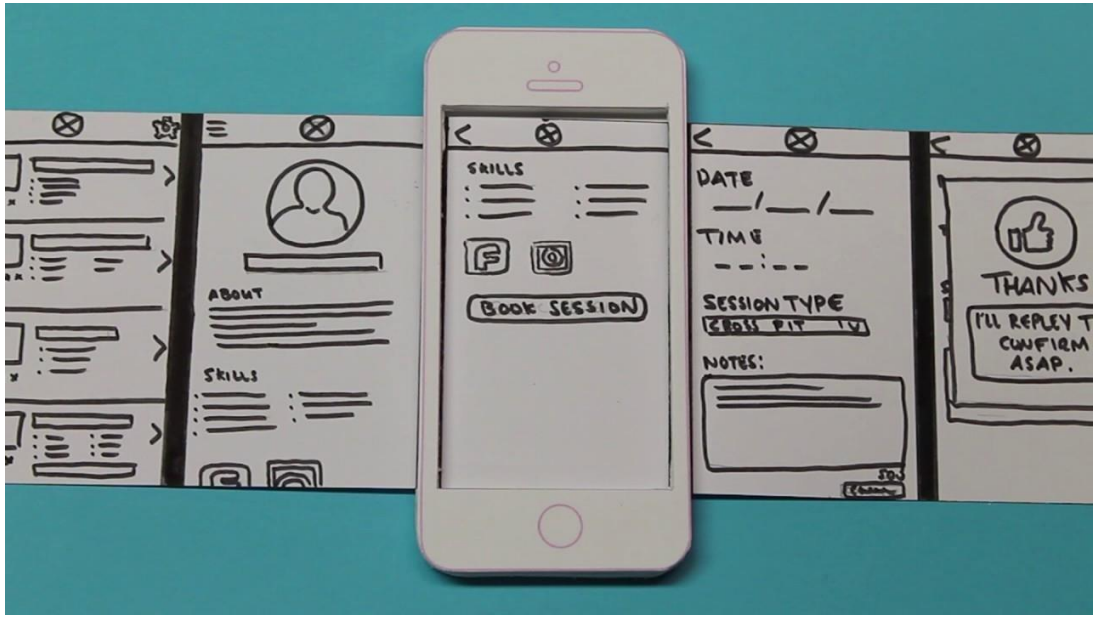
Add sub-part of UI

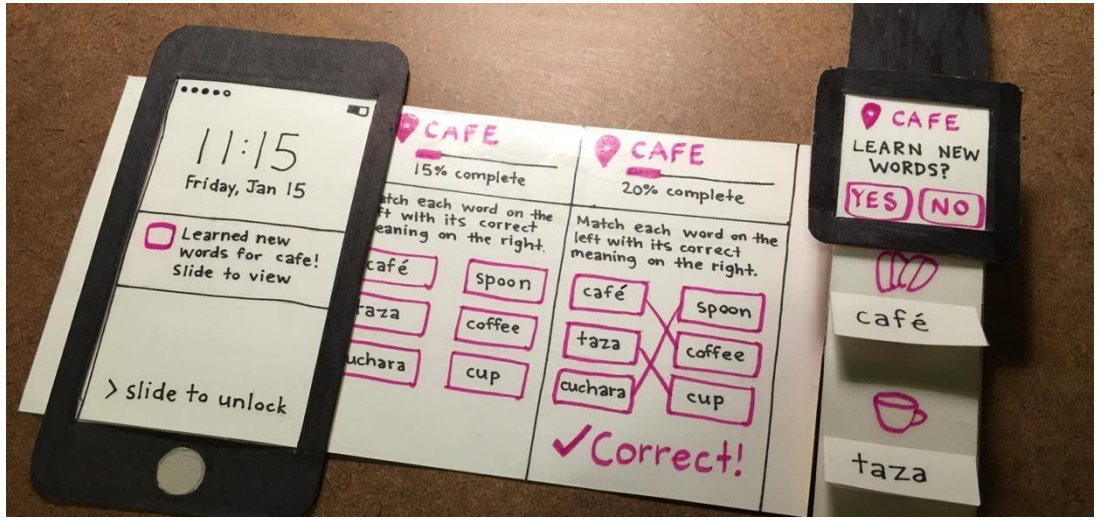


https://youtu.be/GrV2SZuRPvo

“Dynamic” Screens

This is more rigid because you define the sequence but if the user doesn't do the expected one step, you don't go to next screen





How to Test a Paper Prototype

ognuno fa tutte le parti.
Lo si fa internamente
nel nostro caso

- The Design Team should cover these roles
- 'Computer' actor
  - Simulates prototype
  - Does not give any feedback that the computer would not
- Facilitator
  - Presents interface and tasks to the user
  - Encourages user to "**think aloud**" while he/she interact by asking questions
  - Keeps user test from getting off track
- Observer
  - Keeps mouth shut
  - Takes copious notes

Learnable Lessons From Paper Prototypes

- **Can Learn**

  - Conceptual model
    - Do users understand ~~it~~? (the interactions)
  - Functionality
    - Does it do what's needed? Missing features? (by user)
  - Navigation & task flow
    - Can users find their way around?
    - Are information preconditions met?
  - Terminology
    - Do users understand labels?
  - Screen contents
    - What needs to go on the screen?

    During lowfi, not so much stuff on the screen

- **Cannot Learn**

  - Look: color, font, whitespace, etc.
  - Feel: efficiency issues
  - Response time
  - Are small changes noticed? (small changes are too visible)
    - Even the tiniest change to a paper prototype is clearly visible to user
  - Exploration vs. deliberation (sequence completely defined)
    - Users are more deliberate with a paper prototype; they don't explore or thrash as much

References and Acknowledgments

CC BY NC SA

- Google, Begin Today With Rapid prototyping, https://www.youtube.com/playlist?list=PL9KVIdeJ2K8NDpsiyYpcbB_qifd3y5CYZ
- MIT, http://web.mit.edu/6.813/www/sp18/classes/11-prototyping/#reading_11_prototyping
- Scott Klemmer, Storyboards, Paper Prototypes, and Mockups, https://youtu.be/z4glsttyxw8

- These slides are a modified version of the ones of prof. Luigi De Russis, PoliTo.

  A big thanks to Luigi De Russis (PoliTo) for his work!